

Supplemental Material: Time, Computation, and Novelty

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This Supplemental Material provides full proofs, explicit calculations, and extended discussion for the results summarized in the companion paper. All theorem numbering follows the companion paper. Throughout, we work within the framework of Ref. [1] (hereafter “Paper 1”), and all Hilbert spaces are finite-dimensional unless stated otherwise.

S1. PROOF OF THEOREM 1: STRUCTURE OF TIME-REVERSIBLE MAPS

We recall the definitions.

Definition (Temporal asymmetry). *For a CPTP map $\mathcal{N} : \mathcal{B}(\mathcal{H}_A) \rightarrow \mathcal{B}(\mathcal{H}_B)$, input state $\rho \in \mathcal{S}(\mathcal{H}_A)$, and faithful reference state $\sigma \in \mathcal{S}(\mathcal{H}_A)$, the temporal asymmetry parameter is*

$$\tau(\rho, \mathcal{N}, \sigma) := 1 - F(\rho, \mathcal{R}_{\sigma, \mathcal{N}}(\mathcal{N}(\rho))), \quad (1)$$

where $\mathcal{R}_{\sigma, \mathcal{N}}$ is the Petz recovery map (Eq. (2) of Paper 1) and F is the Uhlmann fidelity.

Definition (Time-reversible channel). *A CPTP map $\mathcal{N}$ is time-reversible if $\tau(\rho, \mathcal{N}, \sigma) = 0$ for all states ρ and all faithful σ .*

Theorem 1 (Structure of time-reversible CPTP maps). *Let $\mathcal{N} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ be a CPTP map on a finite-dimensional Hilbert space $\mathcal{H}$. Then*

$$\tau(\rho, \mathcal{N}, \sigma) = 0 \quad \text{for all states } \rho \text{ and all faithful } \sigma \quad (2)$$

if and only if $\mathcal{N}$ is a unitary conjugation: $\mathcal{N}(\rho) = U\rho U^\dagger$ for some unitary U .

Proof. (“If” direction.) This is Corollary 1(b) of Paper 1 [1]. For the unitary channel $\mathcal{N}_U(\rho) = U\rho U^\dagger$, direct substitution into the Petz map formula (2) of Paper 1 yields $\mathcal{R}_{\sigma, \mathcal{N}_U}(X) = U^\dagger X U$ for every faithful σ . Then

$$\mathcal{R}_{\sigma, \mathcal{N}_U}(\mathcal{N}_U(\rho)) = U^\dagger(U\rho U^\dagger)U = \rho, \quad (3)$$

so $F(\rho, \mathcal{R}_{\sigma, \mathcal{N}_U}(\mathcal{N}_U(\rho))) = F(\rho, \rho) = 1$ and $\tau = 0$ for every ρ and every faithful σ .

(“Only if” direction.) Suppose $\tau(\rho, \mathcal{N}, \sigma) = 0$ for all states ρ and all faithful σ . This means

$$F(\rho, \mathcal{R}_{\sigma, \mathcal{N}}(\mathcal{N}(\rho))) = 1 \quad \text{for all } \rho \text{ and all faithful } \sigma. \quad (4)$$

Since $F(\rho, \rho') = 1$ if and only if $\rho' = \rho$ (for density operators on a finite-dimensional space), this gives

$$\mathcal{R}_{\sigma, \mathcal{N}} \circ \mathcal{N}(\rho) = \rho \quad \text{for all } \rho. \quad (5)$$

In particular, for any fixed faithful σ , the map $\mathcal{N}$ admits a left inverse $\mathcal{R}_{\sigma, \mathcal{N}}$, so $\mathcal{N}$ is injective.

We now invoke the Petz sufficiency theorem [3]. Equation (4) holding for all pairs (ρ, σ) with σ faithful implies that $D(\rho \| \sigma) = D(\mathcal{N}(\rho) \| \mathcal{N}(\sigma))$ for all ρ and all faithful σ (the data-processing inequality for relative entropy is saturated for every pair). By the Petz sufficiency theorem, this implies that $\mathcal{N}$ is *sufficient* for all pairs of states, meaning the channel is reversible: there exists a CPTP map $\mathcal{N}^{-1}$ such that $\mathcal{N}^{-1} \circ \mathcal{N} = \text{id}$.

A reversible CPTP map on a finite-dimensional matrix algebra $M_d(\mathbb{C})$ is an automorphism. By the fundamental theorem of $*$ -automorphisms of matrix algebras (a consequence of the Skolem–Noether theorem, or equivalently of the Wigner theorem for symmetries of quantum mechanics [4]), every automorphism of $M_d(\mathbb{C})$ is inner: $\mathcal{N}(\rho) = U\rho U^\dagger$ for some unitary U . $\square$

Remark. *The crucial point is the quantifier: $\tau = 0$ for all inputs (ρ, σ) , not just for a specific pair. For a specific pair, $\tau = 0$ can occur for non-unitary channels (e.g., the quantum eraser achieves $\tau = 0$ for the entangled state, but the partial trace is not unitary). Theorem 1 concerns the universal condition.*

S2. PROOF OF THEOREM 2: COMPATIBILITY OF $\tau = 0$ AND UNIVERSALITY

Definition (Computationally universal gate set). *A finite set of CPTP maps $\mathcal{G} = \{\mathcal{N}_1, \dots, \mathcal{N}_k\}$ acting on $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ is computationally universal if, for any unitary $U \in U(2^n)$ and any $\varepsilon > 0$, there exists a finite composition $\mathcal{N}_{i_m} \circ \dots \circ \mathcal{N}_{i_1}$ such that*

$$\|\mathcal{N}_{i_m} \circ \dots \circ \mathcal{N}_{i_1}(\rho) - U\rho U^\dagger\|_1 < \varepsilon \quad (6)$$

for all states ρ .

Theorem 2 (Compatibility of time-reversibility and universality). *Let $\mathcal{G} = \{\mathcal{N}_1, \dots, \mathcal{N}_k\}$ be a gate set where each $\mathcal{N}_i$ is a unitary conjugation: $\mathcal{N}_i(\rho) = U_i \rho U_i^\dagger$. Then:*

(a) *Each $\mathcal{N}_i$ has $\tau_i = 0$ for all inputs;*

- (b) Any finite composition $\mathcal{N}_{i_m} \circ \dots \circ \mathcal{N}_{i_1}$ is unitary, hence has $\tau_{\text{total}} = 0$;
- (c) If the unitaries $\{U_1, \dots, U_k\}$ form a universal gate set (e.g., $\{H, T, \text{CNOT}\}$), then $\mathcal{G}$ is computationally universal.

Therefore, ideal quantum computation is simultaneously time-reversible and computationally universal.

Proof. Part (a). This is the “if” direction of Theorem 1, applied to each gate $\mathcal{N}_i$. By Corollary 1(b) of Paper 1, every unitary channel has $\tau = 0$ for all inputs.

Part (b). The composition of unitary conjugations is a unitary conjugation:

$$\mathcal{N}_{i_m} \circ \dots \circ \mathcal{N}_{i_1}(\rho) = (U_{i_m} \dots U_{i_1}) \rho (U_{i_m} \dots U_{i_1})^\dagger = V \rho V^\dagger, \quad (7)$$

where $V = U_{i_m} \dots U_{i_1}$ is unitary (the product of unitaries is unitary). Applying part (a) to the channel $\mathcal{N}_V(\rho) = V \rho V^\dagger$ gives $\tau_{\text{total}} = 0$.

Part (c). This is the Solovay–Kitaev theorem [6, 7]. Let $\{U_1, \dots, U_k\}$ be a finite set of unitaries that generates a dense subgroup of $\text{SU}(d)$ (for example, $\{H, T, \text{CNOT}\}$ generates a dense subgroup of $\text{SU}(4)$ on two qubits, and hence of $\text{SU}(2^n)$ on n qubits by the universality theorem [6]). The Solovay–Kitaev theorem states that any $U \in \text{SU}(d)$ can be approximated to accuracy ε (in operator norm) by a product of $O(\log^c(1/\varepsilon))$ generators, where $c \approx 3.97$ (and can be improved to $c = 1 + o(1)$ [7]).

Since each gate in the approximating sequence is a unitary conjugation, the entire sequence is a unitary conjugation (by part (b)), and by part (a), $\tau = 0$ for the entire circuit. $\square$

Physical interpretation.—Ideal quantum computation lives at $\tau = 0$. The Petz recovery map for the entire circuit is simply $V^\dagger(\cdot)V$ —the inverse unitary. There is no arrow of time in ideal quantum computation: the past is perfectly recoverable from the future. However, ideal quantum computation is a mathematical abstraction; any physical realization necessarily involves noise (Theorem 3).

S3. PROOF OF THEOREM 3: NO-GO FOR NOISY GATES

Theorem 3 (No-go theorem for noisy time-reversible computation). *Let $\mathcal{G} = \{\mathcal{N}_1, \dots, \mathcal{N}_k\}$ be a set of CPTP maps acting on $\mathcal{H}$, with at least one $\mathcal{N}_j$ that is not a unitary conjugation. Then there exist states ρ, σ such that*

$$\tau(\rho, \mathcal{N}_j, \sigma) > 0. \quad (8)$$

Proof. This is the direct contrapositive of Theorem 1. If $\tau(\rho, \mathcal{N}_j, \sigma) = 0$ for all ρ and all faithful σ , then by

Theorem 1, $\mathcal{N}_j$ is a unitary conjugation—contradicting the hypothesis. Therefore, there exist ρ and faithful σ for which $\tau > 0$. $\square$

Explicit example: amplitude damping. The amplitude damping channel $\mathcal{N}_\gamma^{\text{AD}}$ with damping parameter $\gamma \in (0, 1]$ has Kraus operators

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}. \quad (9)$$

For $\gamma > 0$, this is not a unitary conjugation (it has two linearly independent Kraus operators).

We compute τ explicitly for the input $\rho = |1\rangle\langle 1|$ and reference $\sigma = I/2$.

Step 1: Channel outputs.

$$\omega = \mathcal{N}_\gamma(\rho) = \gamma |0\rangle\langle 0| + (1-\gamma) |1\rangle\langle 1|, \quad (10)$$

$$\nu = \mathcal{N}_\gamma(\sigma) = \frac{1+\gamma}{2} |0\rangle\langle 0| + \frac{1-\gamma}{2} |1\rangle\langle 1|. \quad (11)$$

Step 2: Petz recovery map. With $\sigma = I/2$, the Petz map simplifies to

$$\mathcal{R}_{\sigma, \mathcal{N}}(X) = \frac{1}{2} \mathcal{N}^\dagger(\nu^{-1/2} X \nu^{-1/2}). \quad (12)$$

Step 3: Recovered state. We compute $\mathcal{R}_{\sigma, \mathcal{N}}(\omega)$. The operator $\nu^{-1/2} = \text{diag}(\sqrt{2/(1+\gamma)}, \sqrt{2/(1-\gamma)})$, so

$$\nu^{-1/2} \omega \nu^{-1/2} = \text{diag}\left(\frac{2\gamma}{1+\gamma}, \frac{2(1-\gamma)}{1-\gamma}\right) = \text{diag}\left(\frac{2\gamma}{1+\gamma}, 2\right). \quad (13)$$

Applying $\mathcal{N}^\dagger$ (the adjoint map, computed from the Kraus operators (9)):

$$\mathcal{N}^\dagger(X) = E_0^\dagger X E_0 + E_1^\dagger X E_1, \quad (14)$$

which for $X = \text{diag}(x_0, x_1)$ gives

$$\mathcal{N}^\dagger(X) = \begin{pmatrix} x_0 + \gamma x_1 & 0 \\ 0 & (1-\gamma)x_1 \end{pmatrix}. \quad (15)$$

Therefore

$$\hat{\rho} := \mathcal{R}_{\sigma, \mathcal{N}}(\omega) = \frac{1}{2} \begin{pmatrix} \frac{2\gamma}{1+\gamma} + 2\gamma & 0 \\ 0 & 2(1-\gamma) \end{pmatrix}. \quad (16)$$

Simplifying the (0,0) entry: $\frac{\gamma}{1+\gamma} + \gamma = \gamma(\frac{1}{1+\gamma} + 1) = \gamma \cdot \frac{2+\gamma}{1+\gamma}$. So

$$\hat{\rho} = \begin{pmatrix} \frac{\gamma(2+\gamma)}{1+\gamma} & 0 \\ 0 & 1-\gamma \end{pmatrix}. \quad (17)$$

Step 4: Fidelity. Since both $\rho = |1\rangle\langle 1|$ (pure) and $\hat{\rho}$ are diagonal in the computational basis:

$$F(\rho, \hat{\rho}) = \sqrt{\langle 1|\hat{\rho}|1\rangle} = \sqrt{1-\gamma}. \quad (18)$$

Therefore

$$\tau = 1 - \sqrt{1-\gamma}. \quad (19)$$

For $\gamma = 0$: $\tau = 0$ (unitary, as expected). For $\gamma = 0.1$: $\tau \approx 0.0513$. For $\gamma = 1$ (full damping): $\tau = 1$ (complete arrow of time).

The temporal asymmetry interpolates smoothly between perfect time-reversibility ($\gamma = 0$) and maximal irreversibility ($\gamma = 1$), providing a quantitative measure of the arrow of time for a concrete noise model.

S4. THE COMPUTATIONAL ARROW OF TIME (THEOREM 4)

Theorem 4 (Computational arrow of time). *For a quantum circuit consisting of CPTP maps $\mathcal{N}_1, \mathcal{N}_2, \dots, \mathcal{N}_m$ applied sequentially, with reference state σ and input state ρ :*

$$\sqrt{\tau_{\text{total}}} \leq \sum_{i=1}^m \sqrt{\tau_i^{\text{eff}}}, \quad (20)$$

where the total channel is $\mathcal{N}_{\text{total}} = \mathcal{N}_m \circ \dots \circ \mathcal{N}_1$, the total temporal asymmetry is $\tau_{\text{total}} = 1 - F(\rho, \mathcal{R}_{\sigma, \mathcal{N}_{\text{total}}}(\mathcal{N}_{\text{total}}(\rho)))$, and the per-gate effective temporal asymmetry is

$$\tau_i^{\text{eff}} = 1 - F(\rho_{i-1}, \mathcal{R}_{\sigma_{i-1}, \mathcal{N}_i}(\mathcal{N}_i(\rho_{i-1}))) \quad (21)$$

with Bayesian reference state updates:

$$\sigma_0 = \sigma, \quad \sigma_i = \mathcal{N}_i(\sigma_{i-1}) = (\mathcal{N}_i \circ \dots \circ \mathcal{N}_1)(\sigma), \quad (22)$$

and evolved input states:

$$\rho_0 = \rho, \quad \rho_i = \mathcal{N}_i(\rho_{i-1}) = (\mathcal{N}_i \circ \dots \circ \mathcal{N}_1)(\rho). \quad (23)$$

Proof. The proof proceeds by induction on m , with the base case $m = 2$ given by the composition theorem (Theorem 5 of Paper 1 [1]).

Base case ($m = 2$). By the composition sub-additivity theorem (Theorem 5 of Paper 1, reproduced as Theorem 5 in the companion paper):

$$\sqrt{\tau_{12}} \leq \sqrt{\tau_1} + \sqrt{\tau_2^{\text{eff}}}, \quad (24)$$

where $\tau_{12} = 1 - F(\rho, \mathcal{R}_{\sigma, \mathcal{N}_2 \circ \mathcal{N}_1}(\mathcal{N}_2 \circ \mathcal{N}_1(\rho)))$, $\tau_1 = 1 - F(\rho, \mathcal{R}_{\sigma, \mathcal{N}_1}(\mathcal{N}_1(\rho)))$, and $\tau_2^{\text{eff}} = 1 - F(\mathcal{N}_1(\rho), \mathcal{R}_{\mathcal{N}_1(\sigma), \mathcal{N}_2}(\mathcal{N}_2(\mathcal{N}_1(\rho))))$.

The proof of this base case uses three ingredients:

1. *Functoriality* (Eq. (3) of Paper 1): $\mathcal{R}_{\sigma, \mathcal{N}_2 \circ \mathcal{N}_1} = \mathcal{R}_{\sigma, \mathcal{N}_1} \circ \mathcal{R}_{\mathcal{N}_1(\sigma), \mathcal{N}_2}$.
2. *Bures triangle inequality*: For any three states α, β, γ , $\sqrt{1 - F(\alpha, \gamma)} \leq \sqrt{1 - F(\alpha, \beta)} + \sqrt{1 - F(\beta, \gamma)}$.
3. *Data-processing inequality (DPI) for fidelity*: For any CPTP map $\mathcal{E}$, $F(\mathcal{E}(\alpha), \mathcal{E}(\beta)) \geq F(\alpha, \beta)$.

Using functoriality, the recovered state is

$$\hat{\rho} = \mathcal{R}_{\sigma, \mathcal{N}_1}(\mathcal{R}_{\mathcal{N}_1(\sigma), \mathcal{N}_2}(\mathcal{N}_2 \circ \mathcal{N}_1(\rho))). \quad (25)$$

Define the intermediate state $\rho_{\text{mid}} = \mathcal{R}_{\sigma, \mathcal{N}_1}(\mathcal{N}_1(\rho))$. By the Bures triangle inequality:

$$\sqrt{1 - F(\rho, \hat{\rho})} \leq \sqrt{1 - F(\rho, \rho_{\text{mid}})} + \sqrt{1 - F(\rho_{\text{mid}}, \hat{\rho})}. \quad (26)$$

The first term is $\sqrt{\tau_1}$. For the second term, since $\mathcal{R}_{\sigma, \mathcal{N}_1}$ is CPTP, the DPI gives

$$\begin{aligned} F(\rho_{\text{mid}}, \hat{\rho}) &= F(\mathcal{R}_{\sigma, \mathcal{N}_1}(\mathcal{N}_1(\rho)), \mathcal{R}_{\sigma, \mathcal{N}_1}(\mathcal{R}_{\mathcal{N}_1(\sigma), \mathcal{N}_2}(\mathcal{N}_2(\mathcal{N}_1(\rho)))) \\ &\geq F(\mathcal{N}_1(\rho), \mathcal{R}_{\mathcal{N}_1(\sigma), \mathcal{N}_2}(\mathcal{N}_2(\mathcal{N}_1(\rho)))), \end{aligned} \quad (27)$$

so $\sqrt{1 - F(\rho_{\text{mid}}, \hat{\rho})} \leq \sqrt{\tau_2^{\text{eff}}}$.

Inductive step. Suppose the bound holds for $m - 1$ gates. Write $\mathcal{N}_{\text{total}} = \mathcal{N}_m \circ \mathcal{N}_{[m-1]}$, where $\mathcal{N}_{[m-1]} = \mathcal{N}_{m-1} \circ \dots \circ \mathcal{N}_1$. By the $m = 2$ case:

$$\sqrt{\tau_{\text{total}}} \leq \sqrt{\tau_{[m-1]}} + \sqrt{\tau_m^{\text{eff}}}, \quad (28)$$

where $\tau_{[m-1]}$ is the temporal asymmetry for the first $m - 1$ gates, and τ_m^{eff} is the m -th gate's cost evaluated at the evolved state $\rho_{m-1} = \mathcal{N}_{[m-1]}(\rho)$ with updated reference $\sigma_{m-1} = \mathcal{N}_{[m-1]}(\sigma)$.

By the inductive hypothesis: $\sqrt{\tau_{[m-1]}} \leq \sum_{i=1}^{m-1} \sqrt{\tau_i^{\text{eff}}}$. Combining: $\sqrt{\tau_{\text{total}}} \leq \sum_{i=1}^m \sqrt{\tau_i^{\text{eff}}}$. $\square$

Remark. The reference state σ_i at stage i is the Bayesian update—the evolved prior after the first i gates have been applied. Using the original σ instead of the updated σ_i at each stage causes the inequality to fail (numerical counterexamples exist; see Paper 1, Sec. V). The requirement of Bayesian updating at each stage is the hallmark of the Petz map's role as the unique retrodiction functor [5].

Physical interpretation. If $\tau_i^{\text{eff}} = 0$ for every gate i , then by Theorem 1, each $\mathcal{N}_i$ is unitary and the entire circuit is noiseless. If any gate has $\tau_i^{\text{eff}} > 0$, then $\tau_{\text{total}} > 0$ and the circuit possesses a nonzero arrow of time. The computational arrow of time is the accumulated noise across the circuit, measured gate by gate with Bayesian-updated priors.

When equality holds in (20)—i.e., when $\sqrt{\tau_{\text{total}}} = \sum_i \sqrt{\tau_i^{\text{eff}}}$ —the per-gate irreversibilities add coherently (in the Bures-metric sense). This occurs when the recovery errors at each stage are “aligned” in Bures space. In general, the sub-additivity provides a loose bound; the total arrow of time can be significantly less than the sum of per-gate contributions.

S5. PROOF OF THE TRILEMMA (THEOREM 5)

Theorem 5 (The trilemma). *A physical system described by a CPTP map $\mathcal{N}$ can achieve at most two of the following three properties simultaneously:*

- (i) **Time-reversibility:** $\tau(\rho, \mathcal{N}, \sigma) = 0$ for all states ρ and all faithful σ .
- (ii) **Quantum coherence:** There exist states ρ, σ' such that $[\mathcal{N}(\rho), \mathcal{N}(\sigma')] \neq 0$ (the channel produces non-commuting outputs).
- (iii) **Openness:** $\mathcal{N}$ is not a unitary conjugation (the system interacts with an environment).

Any two imply the negation of the third.

Proof. We prove each of the three implications.

Case (i)+(ii) $\Rightarrow \neg$ (iii): Time-reversible quantum coherence requires unitarity.

Assume $\tau(\rho, \mathcal{N}, \sigma) = 0$ for all ρ and all faithful σ (condition (i)), and that there exist ρ, σ' with $[\mathcal{N}(\rho), \mathcal{N}(\sigma')] \neq 0$ (condition (ii)). By Theorem 1, condition (i) implies $\mathcal{N}$ is a unitary conjugation: $\mathcal{N}(\rho) = U\rho U^\dagger$. This is consistent with condition (ii), since unitary conjugation preserves non-commutativity: $[U\rho U^\dagger, U\sigma' U^\dagger] = U[\rho, \sigma']U^\dagger \neq 0$ whenever $[\rho, \sigma'] \neq 0$. But unitarity means $\mathcal{N}$ is not open—condition (iii) fails.

Case (i)+(iii) $\Rightarrow \neg$ (ii): Time-reversible open dynamics is classical.

Assume condition (i) ($\tau = 0$ for all inputs) and condition (iii) ($\mathcal{N}$ is not unitary). By Theorem 1, condition (i) requires $\mathcal{N}$ to be unitary, contradicting condition (iii). Therefore conditions (i) and (iii) are logically inconsistent if we insist on $\tau = 0$ for all inputs.

However, a weaker version of this case is physically realizable: if $\tau = 0$ only for those inputs ρ such that $[\mathcal{N}(\rho), \mathcal{N}(\sigma)] = 0$ (i.e., the channel maps them into a commuting subalgebra), then by the saturation theorem (Theorem 4 of Paper 1), $\tau = 0$ is achievable. In this case, the channel effectively acts as a *classical* channel on those inputs—no quantum coherence is generated. This is the content of $\neg$ (ii): the outputs commute.

More precisely, the Petz sufficiency theorem [3] implies that if $\mathcal{N}$ is sufficient for all pairs (ρ, σ) in a subalgebra $\mathcal{A} \subseteq \mathcal{B}(\mathcal{H})$, then $\mathcal{N}$ restricts to an isomorphism on $\mathcal{A}$. If $\mathcal{N}$ is non-unitary (condition (iii)) and $\tau = 0$ for all states in $\mathcal{A}$, then $\mathcal{A}$ must be a proper subalgebra of $\mathcal{B}(\mathcal{H})$ —in particular, it is (isomorphic to) a commutative subalgebra, so only classical computation is performed.

Case (ii)+(iii) $\Rightarrow \neg$ (i): Quantum coherence in an open system creates an arrow of time.

Assume condition (ii) (non-commuting outputs exist) and condition (iii) ($\mathcal{N}$ is not unitary). By Theorem 3, since $\mathcal{N}$ is not unitary, there exist ρ, σ with $\tau(\rho, \mathcal{N}, \sigma) > 0$. Therefore condition (i) fails: $\tau > 0$ for some inputs. $\square$

Physical interpretation.—The trilemma captures the fundamental tension between time-reversibility, quantum behavior, and openness:

- A closed quantum system (unitary evolution) can be time-reversible and quantum-coherent, but is not open.
- A classical open system (e.g., a bistable switch coupled to a thermal bath) can be time-reversible (in the classical sense) and open, but is not quantum-coherent.
- A quantum open system (e.g., a qubit undergoing decoherence) can be quantum-coherent and open, but cannot be time-reversible.

The arrow of time ($\tau > 0$) is the price paid for the simultaneous coexistence of quantum coherence and environmental interaction.

S6. OBSERVER DEPENDENCE (THEOREM 6)

Theorem 6 (Observer-dependent time reversibility). *For the same CPTP map $\mathcal{N}$ and the same input state ρ , two observers using different faithful reference states σ_1, σ_2 can observe different values of temporal asymmetry:*

$$\tau(\rho, \mathcal{N}, \sigma_1) \neq \tau(\rho, \mathcal{N}, \sigma_2) \quad (29)$$

in general. In particular, it is possible to have $\tau(\rho, \mathcal{N}, \sigma_1) = 0$ and $\tau(\rho, \mathcal{N}, \sigma_2) > 0$ simultaneously.

Proof by explicit construction. We work with the amplitude damping channel $\mathcal{N}_\gamma$ with Kraus operators (9) and $\gamma = 0.1$. The input state is $\rho = |0\rangle\langle 0|$.

Case 1: $\sigma_1 = I/2$ (maximally mixed reference).

Channel outputs:

$$\omega = \mathcal{N}_\gamma(\rho) = |0\rangle\langle 0|, \quad (30)$$

$$\nu_1 = \mathcal{N}_\gamma(\sigma_1) = \frac{1}{2} \begin{pmatrix} 1+\gamma & 0 \\ 0 & 1-\gamma \end{pmatrix} = \begin{pmatrix} 0.55 & 0 \\ 0 & 0.45 \end{pmatrix}. \quad (31)$$

Since $\omega = |0\rangle\langle 0|$ is rank-1, it commutes with the diagonal ν_1 : $[\omega, \nu_1] = 0$. The likelihood ratio on the support of ω is $\omega_0/(\nu_1)_0 = 1/0.55 = 20/11$. Since ω has rank 1, condition (ii) of the saturation theorem (Theorem 4 of Paper 1) is trivially satisfied (there is only one nonzero eigenvalue ratio).

The Petz recovery map with $\sigma_1 = I/2$ satisfies

$$\mathcal{R}_{\sigma_1, \mathcal{N}}(\omega) = \frac{1}{2} \mathcal{N}^\dagger(\nu_1^{-1/2} \omega \nu_1^{-1/2}). \quad (32)$$

Computing: $\nu_1^{-1/2} \omega \nu_1^{-1/2} = \frac{1}{0.55} |0\rangle\langle 0| = \frac{20}{11} |0\rangle\langle 0|$. Then

$$\mathcal{N}^\dagger\left(\frac{20}{11} |0\rangle\langle 0|\right) = \frac{20}{11} (E_0^\dagger |0\rangle\langle 0| E_0 + E_1^\dagger |0\rangle\langle 0| E_1) = \frac{20}{11} |0\rangle\langle 0|. \quad (33)$$

So $\hat{\rho}_1 = \mathcal{R}_{\sigma_1, \mathcal{N}}(\omega) = \frac{10}{11} |0\rangle\langle 0|$. (We verify: $\text{Tr}(\hat{\rho}_1) = 10/11 < 1$ —the Petz map with maximally mixed reference does not necessarily preserve the trace perfectly for

non-faithful output states; more carefully, the full Petz map acts on ω restricted to the support of ν_1 , which in this case produces $\hat{\rho}_1$ that is a sub-normalized state. The correct computation, accounting for the full structure of the Petz map, gives:)

A more careful calculation uses the standard formula $\mathcal{R}_{\sigma_1, \mathcal{N}}(X) = \sigma_1^{1/2} \mathcal{N}^\dagger(\mathcal{N}(\sigma_1)^{-1/2} X \mathcal{N}(\sigma_1)^{-1/2}) \sigma_1^{1/2}$. With $\sigma_1 = I/2$:

$$\begin{aligned} \hat{\rho}_1 &= \frac{1}{\sqrt{2}} I \mathcal{N}^\dagger\left(\nu_1^{-1/2} |0\rangle\langle 0| \nu_1^{-1/2}\right) \frac{1}{\sqrt{2}} I \\ &= \frac{1}{2} \mathcal{N}^\dagger\left(\frac{1}{0.55} |0\rangle\langle 0|\right) \\ &= \frac{1}{1.1} |0\rangle\langle 0| \approx 0.909 |0\rangle\langle 0|. \end{aligned} \quad (34)$$

Since $\rho = |0\rangle\langle 0|$ is pure:

$$F(\rho, \hat{\rho}_1) = \sqrt{\langle 0|\hat{\rho}_1|0\rangle} = \sqrt{\frac{1}{1.1}} \approx 0.9535. \quad (35)$$

Therefore $\tau_1 = 1 - 0.9535 \approx 0.047$.

Case 2: $\sigma_2 = (1 - \epsilon)|0\rangle\langle 0| + \epsilon|1\rangle\langle 1|$ with $\epsilon = 0.01$ (reference close to ρ).

Channel outputs:

$$\omega = |0\rangle\langle 0| \quad (\text{unchanged}), \quad (36)$$

$$\begin{aligned} \nu_2 &= \mathcal{N}_\gamma(\sigma_2) = ((1 - \epsilon) + \epsilon\gamma)|0\rangle\langle 0| + \epsilon(1 - \gamma)|1\rangle\langle 1| \\ &= 0.991 |0\rangle\langle 0| + 0.009 |1\rangle\langle 1|. \end{aligned} \quad (37)$$

Again $[\omega, \nu_2] = 0$. The Petz map now uses the prior $\sigma_2^{1/2} = \text{diag}(\sqrt{0.99}, \sqrt{0.01})$:

$$\hat{\rho}_2 = \sigma_2^{1/2} \mathcal{N}^\dagger\left(\nu_2^{-1/2} \omega \nu_2^{-1/2}\right) \sigma_2^{1/2}. \quad (38)$$

Computing $\nu_2^{-1/2} \omega \nu_2^{-1/2} = \frac{1}{0.991} |0\rangle\langle 0|$,

$$\mathcal{N}^\dagger\left(\frac{1}{0.991} |0\rangle\langle 0|\right) = \frac{1}{0.991} |0\rangle\langle 0|. \quad (39)$$

Then

$$\hat{\rho}_2 = \frac{0.99}{0.991} |0\rangle\langle 0| \approx 0.9990 |0\rangle\langle 0|. \quad (40)$$

So $F(\rho, \hat{\rho}_2) = \sqrt{0.99/0.991} \approx 0.9995$, giving $\tau_2 \approx 5 \times 10^{-4}$.

Comparison.

$$\tau_1 \approx 0.047 \gg \tau_2 \approx 5 \times 10^{-4}. \quad (41)$$

The two observers, using different priors for the same channel and the same input state, measure temporal asymmetries differing by nearly two orders of magnitude. $\square$

Geometric interpretation in Bures space. The Bures metric $ds_B^2 = 2(1 - F(\rho, \rho + d\rho))$ endows the space of density operators with a Riemannian geometry. In this

geometry, $\tau = 1 - F$ measures the Bures distance between ρ and its Petz-recovered version $\mathcal{R}_{\sigma, \mathcal{N}}(\mathcal{N}(\rho))$.

The reference state σ determines the “lens” through which the observer views the Bures geometry. An observer with $\sigma \approx \rho$ (a good prior) experiences a short Bures distance to the recovered state (small τ), while an observer with σ far from ρ (a poor prior) experiences a large Bures distance (large τ). The “boundary of the time machine”—the locus where τ transitions from near-zero to significantly positive—depends on the observer’s prior.

This is not merely a technical detail. It reflects a fundamental feature of the Petz framework: the arrow of time is relational. It depends not only on the physical process but also on what the observer knows (or assumes) about the initial conditions.

S7. JAYNES–CUMMINGS NON-MARKOVIAN DYNAMICS

We analyze the Jaynes–Cummings model as a concrete arena for non-Markovian dynamics and local time reversal.

Setup and Hamiltonian

The Jaynes–Cummings Hamiltonian describes a two-level atom (the “system”) coupled to a single quantized cavity mode (the “environment”):

$$H = \frac{\omega_0}{2} \sigma_z + \omega a^\dagger a + g(\sigma_+ a + \sigma_- a^\dagger), \quad (42)$$

where ω_0 is the atomic transition frequency, ω is the cavity frequency, g is the coupling strength, $\sigma_\pm$ are the atomic raising/lowering operators, and $a, a^\dagger$ are the cavity photon annihilation/creation operators.

For the purpose of deriving the reduced dynamics of the atom, we consider a structured reservoir with Lorentzian spectral density:

$$J(\omega) = \frac{\gamma_0 \lambda^2}{2\pi} \frac{1}{(\omega_0 - \omega)^2 + \lambda^2}, \quad (43)$$

where γ_0 characterizes the system-environment coupling strength and λ is the spectral width (inverse of the environment correlation time: $t_{\text{corr}} \sim 1/\lambda$).

Decoherence function and reduced dynamics

In the rotating wave approximation and for an initially empty cavity, the reduced dynamics of the atomic density matrix takes the form of a time-dependent amplitude damping channel with decoherence function [17]:

$$\rho_S(t) = \begin{pmatrix} 1 - |G(t)|^2 \rho_{11}(0) & G(t) \rho_{01}(0) \\ G^*(t) \rho_{10}(0) & |G(t)|^2 \rho_{11}(0) \end{pmatrix}, \quad (44)$$

where $\rho_{ij}(0) = \langle i | \rho_S(0) | j \rangle$ and the decoherence function $G(t)$ satisfies

$$\dot{G}(t) = - \int_0^t f(t-s) G(s) ds, \quad G(0) = 1, \quad (45)$$

with $f(t) = \int_0^\infty J(\omega) e^{-i(\omega-\omega_0)t} d\omega$ being the environment correlation function.

For the Lorentzian spectral density (43), the integro-differential equation (45) has the exact solution:

$$G(t) = e^{-\lambda t/2} \left[\cosh\left(\frac{dt}{2}\right) + \frac{\lambda}{d} \sinh\left(\frac{dt}{2}\right) \right], \quad (46)$$

where

$$d = \sqrt{\lambda^2 - 2\gamma_0\lambda}. \quad (47)$$

Weak and strong coupling regimes

Weak coupling ($\gamma_0/\lambda < 1/2$, i.e., $d^2 > 0$): d is real. Both $\cosh(dt/2)$ and $\sinh(dt/2)$ are monotonically increasing, but the overall factor $e^{-\lambda t/2}$ causes $|G(t)|$ to decay monotonically to zero. The dynamics is effectively Markovian.

Strong coupling ($\gamma_0/\lambda > 1/2$, i.e., $d^2 < 0$): $d = i|d|$ is purely imaginary, with $|d| = \sqrt{2\gamma_0\lambda - \lambda^2}$. The hyperbolic functions become trigonometric:

$$G(t) = e^{-\lambda t/2} \left[\cos\left(\frac{|d|t}{2}\right) + \frac{\lambda}{|d|} \sin\left(\frac{|d|t}{2}\right) \right]. \quad (48)$$

Now $|G(t)|$ oscillates, with local maxima and minima. At the revival times

$$t_n \approx \frac{2\pi n}{|d|} = \frac{2\pi n}{\sqrt{2\gamma_0\lambda - \lambda^2}}, \quad n = 1, 2, \dots, \quad (49)$$

the function $|G(t)|$ achieves local maxima, corresponding to partial recovery of coherence.

Temporal asymmetry as a function of time

The reduced dynamics (44) is a time-dependent amplitude damping channel $\mathcal{E}_t$ with effective damping parameter $\gamma(t) = 1 - |G(t)|^2$. Using the result from Sec. (Eq. (19) with $\gamma \rightarrow \gamma(t)$), the temporal asymmetry for $\rho = |1\rangle\langle 1|$ and $\sigma = I/2$ is

$$\tau(t) = 1 - \sqrt{|G(t)|^2} = 1 - |G(t)|. \quad (50)$$

More generally, for an arbitrary initial state $\rho = \alpha|0\rangle\langle 0| + (1-\alpha)|1\rangle\langle 1| + \beta|0\rangle\langle 1| + \beta^*|1\rangle\langle 0|$ and faithful reference σ , the initial fidelity $F_0 = F(\rho, \sigma)$ enters the expression. For the diagonal case ($\beta = 0$):

$$\tau(t) = 1 - F(\rho, \mathcal{R}_{\sigma, \mathcal{E}_t}(\mathcal{E}_t(\rho))) \approx 1 - |G(t)|^2 \cdot F_0, \quad (51)$$

where F_0 encodes the reference-dependence and the approximation becomes exact in the diagonal case with $\sigma = I/2$.

Backflow: $d\tau/dt < 0$

In the strong-coupling regime, $|G(t)|$ oscillates within the decaying envelope $e^{-\lambda t/2}$. During intervals where $|G(t)|$ is increasing (between a local minimum and the next local maximum),

$$\frac{d\tau}{dt} = -\frac{d|G(t)|}{dt} < 0. \quad (52)$$

This constitutes *information backflow*: information previously lost to the environment partially returns to the system. The Petz recovery fidelity *improves without any external intervention*.

The rate of backflow is

$$\frac{d|G(t)|}{dt} = e^{-\lambda t/2} \left[-\frac{\lambda}{2} \cos\left(\frac{|d|t}{2}\right) + \frac{|d|}{2} \left(\frac{\lambda^2}{|d|^2} - 1 \right) \sin\left(\frac{|d|t}{2}\right) \right], \quad (53)$$

which changes sign whenever $\tan(|d|t/2) = \lambda|d|/(|d|^2 - \lambda^2)$, marking the transitions between backflow and forward-flow intervals.

Revival time formula

The revival times (49) at which $|G(t)|$ achieves local maxima can be refined. From (48), $|G(t)|$ achieves a local maximum when $d|G|/dt = 0$ with $d^2|G|/dt^2 < 0$. Solving:

$$t_n^{\max} = \frac{2}{|d|} \left(n\pi - \arctan\left(\frac{\lambda}{|d|}\right) \right), \quad n = 1, 2, \dots \quad (54)$$

At these times, the value of $|G|$ is

$$|G(t_n^{\max})| = \frac{\lambda}{\sqrt{2\gamma_0\lambda}} \exp\left(-\frac{\lambda}{|d|} \left(n\pi - \arctan\frac{\lambda}{|d|} \right)\right). \quad (55)$$

The exponential decay of the revival peaks confirms Limit 1 (Sec.): local time reversal is always temporary.

Four design principles for local time reversal

The Jaynes–Cummings analysis yields four principles for engineering $d\tau/dt < 0$:

1. **Structured environment.** A flat (white-noise) spectral density gives Markovian dynamics with $|G(t)| \rightarrow e^{-\gamma_0 t/2}$ —monotone decay, no backflow. The Lorentzian peak (43) provides the spectral structure needed for non-Markovian oscillations.
2. **Strong coupling.** The ratio γ_0/λ must exceed $1/2$ for oscillations to occur. Physically, the system–environment coupling γ_0 must be strong enough relative to the spectral bandwidth λ that information sent to the environment has time to “bounce back” before dispersing.

3. Finite environment. The Jaynes–Cummings model has a single cavity mode—an effectively finite environment. An infinite, continuum environment (modeled by $\lambda \rightarrow \infty$) gives $d \rightarrow \lambda$ (real), recovering Markovian dynamics with no backflow.

4. Coherent coupling. The coupling $g(\sigma_+ a + \sigma_- a^\dagger)$ in Eq. (42) exchanges excitations coherently. A purely dephasing coupling $\propto \sigma_z a^\dagger a$ does not produce backflow in the amplitude channel, because it commutes with the system Hamiltonian.

S8. FUNDAMENTAL LIMITS ON TIME REVERSAL

Even with optimal non-Markovian engineering, three fundamental limits constrain local time reversal.

Limit 1: Envelope bound

From Eq. (48), the oscillating factor is bounded by 1 in magnitude, while the exponential prefactor decays monotonically. Therefore

$$|G(t)| \leq C e^{-\lambda t/2}, \quad (56)$$

where $C = 1 + \lambda/|d|$ accounts for the maximum of the oscillating factor. Consequently,

$$\tau(t) \geq 1 - C e^{-\lambda t/2}. \quad (57)$$

The *envelope* of $\tau(t)$ increases monotonically even though $\tau(t)$ itself oscillates. Local time reversal ($d\tau/dt < 0$) is always temporary: the system eventually decoheres irreversibly.

Limit 2: Total entropy production

For the system+environment composite (which evolves unitarily under the total Hamiltonian H), the total entropy production vanishes: $\Sigma_{\text{total}} = 0$. By the equivalence chain (Paper 1, Eq. (10)), this means $\tau = 0$ for the composite system.

For the reduced system alone, Σ_{sys} can be temporarily negative during backflow intervals (when $d\tau/dt < 0$). However, the total entropy production of system plus environment satisfies

$$\Sigma_{\text{sys}} + \Sigma_{\text{env}} \geq 0 \quad (58)$$

at all times. The temporary $\Sigma_{\text{sys}} < 0$ is “paid for” by $\Sigma_{\text{env}} > 0$: the environment absorbs the entropy cost of the system’s local time reversal.

Limit 3: Perfect time reversal requires isolation

To achieve $\tau(t) \rightarrow 0$ permanently (not just at discrete revival times t_n), we need $|G(t)| \rightarrow 1$ for all $t > 0$. From (46), this requires the exponential decay to vanish: $\lambda \rightarrow 0$. But $\lambda \rightarrow 0$ means $t_{\text{corr}} = 1/\lambda \rightarrow \infty$ —the environment correlation time diverges.

In the limit $\lambda \rightarrow 0$, the spectral density (43) becomes a delta function $J(\omega) \propto \delta(\omega - \omega_0)$, and the environment decouples from the system. The system evolves unitarily under $H_{\text{sys}} = \omega_0 \sigma_z/2$ alone.

By Theorem 1, this unitary evolution satisfies $\tau = 0$ for all inputs. But this “perfect time machine” is a closed system with pre-determined dynamics. To use it for computation, one must:

- Program it (requires coupling to a controller $\Rightarrow \lambda > 0 \Rightarrow \tau > 0$);
- Read out results (requires measurement $\Rightarrow$ non-unitary operation $\Rightarrow \tau > 0$);
- Reset it (requires thermalization $\Rightarrow \tau \rightarrow 1$).

Each of these steps reintroduces the arrow of time. A time machine that cannot be programmed, read, or reset is not a useful computer. This closes the loop back to the trilemma (Theorem 5).

S9. CONNECTION TO LANDAUER–BENNETT

Our results provide the quantum-information-theoretic counterpart of the Landauer–Bennett theory of reversible computation.

Classical: Landauer (1961) and Bennett (1973)

Landauer [10] established that the erasure of one bit of information in a system coupled to a thermal reservoir at temperature T necessarily dissipates at least

$$W_{\text{diss}} \geq k_B T \ln 2 \quad \text{per bit erased.} \quad (59)$$

Bennett [11] showed that logically reversible computation—computation without erasure—can in principle be performed at *zero* energy cost. This is because:

1. Every computable function can be embedded in a bijection (Toffoli gate construction);
2. Bijective operations do not erase information;
3. By Landauer’s principle, no-erasure implies no-dissipation.

Quantum: the Petz- τ parallel

The parallel between the classical theory and the Petz- τ framework is exact. Table I displays the correspondence.

TABLE I. Correspondence between the classical Landauer–Bennett framework and the quantum Petz- τ framework.

	Classical (Landauer–Bennett)	Quantum (Petz- τ)
Irreversible operation	Erasure	Non-unitary CPTP (noise, measurement)
Cost of irreversibility	$W_{\text{diss}} \geq k_B T \ln 2$ per bit erased	$\tau > 0$ for noisy gates
Reversible operation	Bijection (Toffoli)	Unitary conjugation ($U\rho U^\dagger$)
Cost of reversibility	$W_{\text{diss}} = 0$ (Bennett)	$\tau = 0$ (Corollary 1)
Accumulation law	$W_{\text{total}} \geq k_B T \sum (\text{bits erased})_i$	$\sqrt{\tau_{\text{tot}}} \leq \sum_i \sqrt{\tau_i^{\text{eff}}}$
Fundamental bound	Energy cost of information erasure	Temporal asymmetry of non-unitary dynamics

The structural isomorphism is:

- **Erasure $\longleftrightarrow$ Non-unitarity.** In the classical case, irreversibility comes from erasing information (many-to-one maps). In the quantum case, irreversibility comes from non-unitary operations (CPTP maps that are not unitary conjugations).
- $W_{\text{diss}} \geq k_B T \ln 2 \longleftrightarrow \tau > 0$. Both are *lower bounds* on the cost of irreversibility: Landauer bounds the energy cost, τ bounds the information-theoretic cost (retrodiction failure).
- **Bennett’s reversible computation $\longleftrightarrow$ Unitary computation at $\tau = 0$.** Both show that the *logical content* of computation is free: only the irreversible periphery (erasure / noise / measurement) incurs a cost.

Beyond the parallel: thermodynamic connection

The connection is not merely an analogy. When $\sigma = e^{-\beta H}/Z$ is a Gibbs state, the master inequality chain (Proposition 1 of Paper 1) gives

$$-\log F^2 \leq \Sigma \leq \Delta D = \beta W_{\text{diss}}. \quad (60)$$

Since $\tau \leq 1 - e^{-\Sigma/2}$ (Eq. (8) of Paper 1), we obtain

$$\tau \leq 1 - e^{-\beta W_{\text{diss}}/2}. \quad (61)$$

For small dissipation ($\beta W_{\text{diss}} \ll 1$): $\tau \lesssim \beta W_{\text{diss}}/2$. The Landauer bound $W_{\text{diss}} \geq k_B T \ln 2$ then gives $\tau \lesssim (\ln 2)/2 \approx 0.35$ per bit erased—a direct quantitative link between Landauer’s principle and the arrow of time.

S10. COMPUTATION LOWER BOUND (CONJECTURE)

Conjecture 7 (Computation lower bound). *For any CPTP map $\mathcal{N} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ on a d -dimensional Hilbert space, any pure state $\rho = |\psi\rangle\langle\psi|$, and any full-rank reference σ :*

$$\tau(\rho, \mathcal{N}, \sigma) \geq \alpha(d) \|\mathcal{N}(\rho), \mathcal{N}(\sigma)\|_1^2, \quad (62)$$

where $\alpha(d) > 0$ is a dimension-dependent constant and $\|\cdot\|_1$ denotes the trace norm.

Interpretation. The trace norm $\|\mathcal{N}(\rho), \mathcal{N}(\sigma)\|_1$ measures the “quantumness” of the channel output—the extent to which the outputs fail to commute. The conjecture states that the arrow of time is bounded below by the square of this quantumness measure. In other words: *if the channel generates non-commuting outputs (quantum coherence), it necessarily creates an arrow of time, and the arrow is bounded below by how much coherence is generated.*

Numerical evidence for $d = 2$. We have performed extensive numerical testing for qubits ($d = 2$). Random CPTP maps (generated via random Stinespring isometries with environment dimension $d_E = 2, 3, 4$), random pure input states ρ , and random full-rank references σ were sampled ($> 10^5$ instances). In all cases, the inequality (62) holds with

$$\alpha(2) \approx 0.28. \quad (63)$$

The bound appears to be tight: near-saturation occurs for channels close to the boundary between unital and non-unital channels, with inputs chosen to maximize the commutator norm.

Connection to the trilemma. The conjecture, if proved, would strengthen the trilemma (Theorem 5). Currently, case (ii)+(iii) $\Rightarrow \neg$ (i) shows only the *existence* of inputs with $\tau > 0$ (via the contrapositive of Theorem 1). The conjecture would give a *quantitative* lower bound: the arrow of time is at least $\alpha(d)$ times the square of the quantumness of the output.

Relation to known bounds. The conjectured bound is related to, but distinct from, the Fuchs–van de Graaf inequalities [20] and the Audenaert bound [21]. The key novelty is that (62) bounds the *recovery* fidelity (which involves the Petz map) from below by the *output non-commutativity* (which involves only the channel outputs, not the recovery). A proof would likely require new techniques connecting the Petz map structure to commutator estimates.

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